

CS671 – Deep Learning and Its Applications

Technical Documentation Report

Model: Unpaired Diffusion (Brain)

Project Group: 5 Region: Brain Modality: Unpaired (CT $\leftrightarrow$ MRI)

1. ARCHITECTURAL OVERVIEW

Unpaired Diffusion (Cycle Diffusion) addresses the challenge of modality translation when spatially aligned pairs are unavailable. It combines the generative power of Latent Diffusion Models (LDM) with the structural constraints of Cycle-Consistency. The model learns to map CT latents to MRI latents by ensuring that the transformation is reversible.

1.1 Cycle-Consistent Latent Diffusion

The architecture integrates a VAE-based latent space with a pair of diffusion-based generators that operate on latent representations.

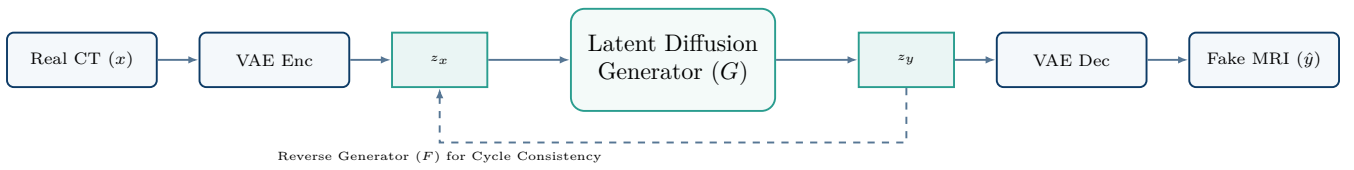


Figure 1: Unpaired Cycle Diffusion: Performing modality translation in the latent feature space.

Latent Representation: By operating in the VAE latent space, the model avoids the high computational cost of pixel-level diffusion while maintaining the ability to reconstruct fine brain structures (e.g., sulci and gyri) during the decoding stage.

Cycle Denoising: The model employs two diffusion pipelines. Generator G translates CT latents to the MRI manifold, while generator F performs the inverse. Cycle consistency is enforced by requiring that $F(G(z_x)) \approx z_x$. This prevents the diffusion process from "drifting" and hallucinating anatomically incorrect structures in the brain.

2. TRAINING DYNAMICS

The training objective combines latent-level adversarial losses with cycle-consistency constraints.

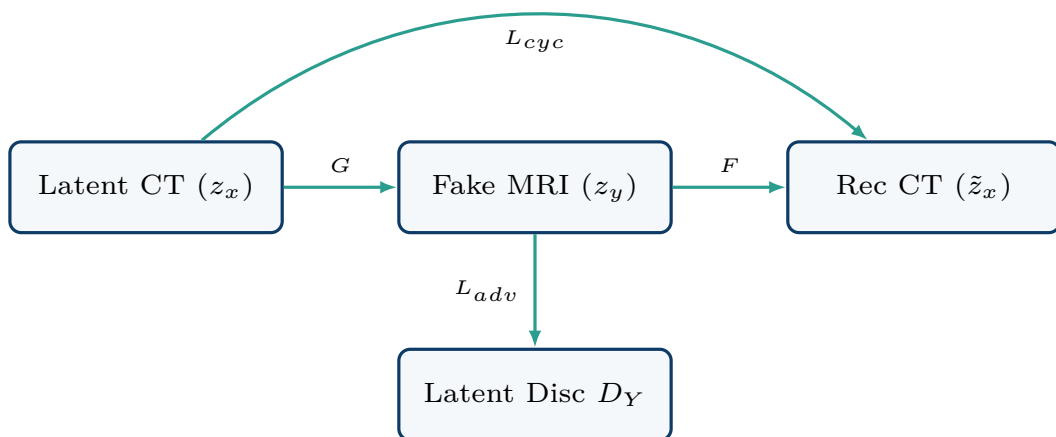


Figure 2: Unpaired Diffusion Training: Adversarial and Cycle-Consistency loops in latent space.

Adversarial Supervision: A latent-space discriminator is trained to distinguish between encoded real MRI latents and generated MRI latents. This ensures that the diffusion output remains on the MRI manifold.

Optimization Stability: Unlike standard GANs, the use of a diffusion generator provides more stable training gradients, especially for the complex, high-contrast textures found in Brain MRI scans.

3. INFERENCE PIPELINE

Inference involves encoding the CT volume, performing latent translation, and decoding the result.



Figure 3: Inference flow: End-to-end latent translation and reconstruction.

The inference engine processes the volume slice-by-slice. The CT latent is passed through the diffusion generator, which performs a set of denoising steps (guided by the CT latent itself) to produce a corresponding MRI latent. This latent is then decoded into the final image space.